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MTH101 - Calculus And Analytical Geometry

If definite integral of f(x)=Sinx over [a,0] is equal to '-2' then the value 'a' is--
---

- pi/2
- ***pi***
- 0
- -pi

If the definite integral of f(x)=3 over [1,x] is greater than '12' then -----

- x>12
- x>5
- ***x>3***
- x>1

If [-8,8] is subdivided into '16' equally spaced subintervals, then the RIGHT end point of 13th sub-interval will be--------.

- 2
- 3
- 4
- ***5***

Which of the following is the integral of sin(2x)?

- cos(2x)+C
- 2cos(2x)+C
- ***–(1/2)cos(2x)+C***
- none

Sum of cubes of 'n' terms of a series whose nth term is 'n' = ---

- Square of n(n+1)(2n+1)/6
- Square of n(n+1)/2
- ***Square of (n+1)/2 (just a guess)***
- Square of n(n+1)/6

In a rectangular coordinate system the coordinate exes divide the plane into four regions called…

- Cube
- ***Quadrants***
- Circle

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An equation of the form is called ….
- Parabolic
- Cubic
- ***Quadratic***

Graph of f(x) =|x| is ……
- Circle shape
- ***V shaped***
- U shaped

Y=f(x) +3 translation of the function is at 3 is ….. By c units
- ***Up***
- Down
- Left
- Right

Graph of y=|x-5| …… of the graph y=|x| 5 units to the right to get the graph of y.
- Reading
- ***Translation***
- Creating
- Definition

By the definition and …… be on the interval (xo, x1)
- ***Greater than zero***
- Equal to zero
- Less than zero
- Less than and equal to zero

Cot(x) discontinuous at …….
- Cos(x) =0
- ***Sin(x) = 0***
- Tan(x) =0
- Cot(x) = 0

If a function is differentiable then the function is also …… on that point
- Discontinues
- No exit
- ***Continuous***

Tan (5-3x) find differentiation w.r.t. x
- ***-3sce² (5-3x)***
- 3sce² (5-3x)

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- 3cos (5-3x)
- 3cos² (5-3x)

Derivative of the will be solved by …….. Method

- ***Implicit***
- Explicit
- Simple

Constant function has the slope …..

- Positive
- Negative
- ***Zero***

Decreasing function has the slope …….

- ***Negative***
- Positive
- Zero

The equation of line of the form is known as

- Point-slope form
- Two points form
- Intercepts form
- ***Slope intercept form***

The set of rational number is a subset of

- ***Integers***
- Natural numbers
- Odd integers
- Real number

If Sin(3x^2) / 6 + C is the anti–derivative of a function f(x), then f(x) = ____.

- x^2 Cos(3x^2).
- ***x Cos(3x^2).***
- x Cos(3x).
- none

Which of the following is the integral of Sin(3x+5) with respect to x ?

- -1/3[-Cos(3x+5)]
- ***1/3[-Cos(3x+5)]***
- 1/15[-Cos(3x+5)]
- -Cos(3x+5)

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If 'n' goes from 1 to 3 and the summation of 'na' = 6a, then the value of 'a' is
---
‣ 6
‣ -6
‣ 1
‣ undetermined

If 'n' goes from 1 to any large ODD number then the summation of '(-1)^n' =

‣-1
‣0
‣1
‣that specific large ODD number

1+2+3.....+t equals
‣n(n+1)/2
‣t(t+1)/2
‣n(n+1)(2n+1)/6
‣none

If definite integral of f(x)=Sinx over [a,0] is equal to '-2' then the value 'a' is--
---
‣pi/2
‣pi
‣0
‣-pi

Question No: 25 (Marks: 3)

Find $\frac{dy}{dx}$ if $y = x^{\frac{p}{q}}$

Question No: 26 (Marks: 5)

Differentiate $\sqrt{13x^2 - 5x + 8}$

Question No: 27 (Marks: 10)

Determine the intervals in which the graph of the function $f(x) = \frac{6}{x^2 + 3}$ is concave upward or downward

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